$^{\mathbf{15}}_{8}\mathrm{O}_{7}$

1 Decay Scheme

^{15}O disintegrates at 99.9% by β^+ emission to the ground state of the stable ^{15}N nuclide.
Le ^{15}O *se désintègre à 99,9 % par émission* β^+ *vers le niveau fondamental du* ^{15}N.

2 Nuclear Data

$T_{1/2}(^{15}\mathrm{O})$	:	122,266	(49)	s
$Q^+(^{15}\mathrm{O})$	:	2754,18	(49)	keV

2.1 Electron Capture Transitions

	Energy (keV)	Probability (%)	Nature	log ft	P_K	P_L
$\epsilon_{0,0}$	2754,18 (49)	0,0999 (17)	Allowed	3,6449 (6)	0,9284 (4)	0,0716 (4)

2.2 β^+ Transitions

	Energy (keV)	Probability (%)	Nature	log ft
$\beta^+_{0,0}$	1732,18 (49)	99,9001 (17)	Allowed	3,6449 (6)

3 Atomic Data

3.1 N

ω_K : 0,0044 (4)

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
$\beta^+_{0,0}$	max:	1732,18 (49)	99,9001 (17)
	avg:	733,47 (23)	

5 Photon Emissions

5.1 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma^{\pm}$	511	199,8002 (34)

6 Main Production Modes

$^{14}N(p,\gamma)^{15}O$

$^{14}N(d,n)^{15}O$

$^{12}C(\alpha,n)^{15}O$

$^{20}Ne(\gamma,n\alpha)^{15}O$
Possible impurities: ^{11}C, ^{19}Ne

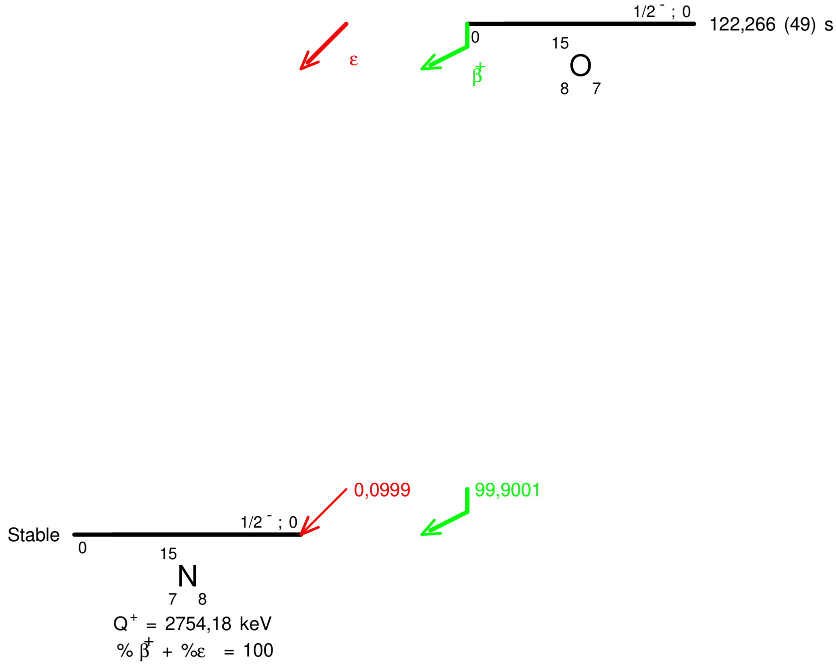
$^{15}N(p,n)^{15}O$

$^{16}O(^{3}He,\alpha\gamma)^{15}O$

7 References

- M.S.Livingston, E.McMillan. Phys. Rev. 46 (1934) 437
  (Half-life, E_{max})
- E.McMillan, M.S.Livingston. Phys. Rev. 47 (1935) 452
  (Half-life)
- W.A.Fowler, L.A.Delsasso, C.C.Lauritsen. Phys. Rev. 49 (1936) 561
  (E_{max})
- W.E.Stephens, K.Djanab, T.W.Bonner. Phys. Rev. 52 (1937) 1079
  (E_{max})
- H.Brown, V.Perez-Mendez. Phys. Rev. 78 (1950) 649
  (Half-life, E_{max})
- R.M.Kline, D.J.Zaffarano. Phys. Rev. 96 (1954) 1620
  (Half-life)
- S.Bashkin, R.R.Carlson, E.B.Nelson. Phys. Rev. 99 (1955) 107
  (Half-life)
- J.D.Kington, J.K.Bair, H.O.Cohn, H.B.Willard. Phys. Rev. 99 (1955) 1393
  (E_{max})

- O.C.Kistner, A.Schawarzschild, B.M.Rustad, D.E.Alburger. Phys. Rev. 105 (1957) 1339
  (Half-life, E_{max})
- J.R.Penning, F.H.Schmidt. Phys. Rev. 105 (1957) 647
  (Half-life)
- O.C.Kistner, B.M.Rustad. Phys. Rev. 114 (1959) 1329
  (Half-life)
- J.Janecke. Z. Naturforsch. 15A (1960) 593
  (Half-life)
- J.Csikai, G.Peto. Phys. Lett. 4 (1963) 252
  (Half-life)
- S.S.Vasil'ev, L.Ya.Shavtvalov. Izv. Akad. Nauk SSSR, Ser. Fiz. 27 (1963) 1261
  (Half-life)
- J.W.Nelson, E.B.Carter, G.E.Mitchel, R.H.Davis. Phys. Rev. 129 (1963) 1723
  (Half-life)
- W.Leiper, R.W.P.Drever. Phys. Rev. C 6 (1972) 1132
  (K/β^+)
- G.Azuelos, J.E.Kitching, K.Ramavataram. Phys. Rev. C 15 (1977) 1847
  (Half-life)
- W.Bambynek, H.Behrens, M.H.Chen, B.Crasemann, M.L.Fitzpatrick, K.W.D.Ledingham, H.Genz, M.Muttere, R.L.Intemann. Rev. Mod. Phys. 49 (1977) 77
  (Theory, Compilation)
- F.Ajzenberg-Selove. Nucl. Phys. A 523 (1991) 1
  (Evaluation)
- E.Schönfeld, H.Janssen. Nucl. Instrum. Methods Phys. Res. A 369 (1996) 527
  (Atomic data)
- C.Dulieu, M.-M.Bé, V.Chisté. Proc. Intern. Conf. Nuclear Data for Science and Technology (2008) p.97
  (Saisinuc)
- X.Mougeot. Appl. Radiat. Isotopes 154 (2019) 108884
  (BetaShape)
- D.P.Burdette, M.Brodeur, D.W.Bardayan, F.D.Becchetti, D.Blankstein, C.Boomershine, L.Caves, S.L.Henderson, J.J.Kolata, B.Liu, J.Long, P.D.O'Malley, S.Y.Strauss. Phys. Rev. C 101 (2020) 055504
  (Half-life)
- M.Wang, W.J.Huang, F.G.Kondev, G.Audi, S.Naimi. Chin. Phys. C 45 (2021) 030003
  (Q-value)
- N.Severijns, L.Hayen, V.De Leebeeck, S.Vanlangendonck, K.Bodek, D.Rozpedzik, I.S.Towner. Phys. Rev. C 107 (2023) 015502
  (Theory, Compilation)
- X.Mougeot. Appl. Radiat. Isotopes 201 (2023) 111018
  (BetaShape)


1/2⁻ ; 0
122,266 (49) s
0
15 O 8 7
ε
β+
0,0999
99,9001
Stable
1/2⁻ ; 0
0
15 N 7 8
Q+ = 2754,18 keV
% β+ + %ε = 100